

Performance Optimization & Benchmark Report: UDIAP Platform

Jina la Mfumo:	User Decision Intelligence Analysis Platform (UDIAP)
Toleo la Mfumo:	v1.0.0-MVP (Production Baseline)
Tarehe ya Usanifu:	Agosti 2026
Mazingira ya Upimaji:	Local Containerized Harness & Cloud Staging (Render + Vercel)
Kiwango cha Document:	PDF-Ready / Enterprise Technical Standard

1. Executive Summary

Taarifa hii inawasilisha uchambuzi wa kina wa utendaji kazi wa mfumo wa **UDIAP** kupitia mbinu za *Site Reliability Engineering (SRE)* na *Systems Performance Profiling*. Lengo kuu la tathmini hii ni kubainisha uwezo wa mfumo chini ya mzigo wa matumizi (load capacity), kutambua vikwazo vya mfumo (bottlenecks), na kuweka viwango halisi vya vipimo (SLIs/SLOs) ili kuandaa mfumo kwa ajili ya usambazaji wa kiwango cha kibiashara.

Matokeo Makuu (Key Findings):

- Synchronous Bottleneck:** Uchakataji wa sasa wa *Cognitive Observer Engine (COE)* na uundaji wa ripoti za AI unachukua wastani wa **3.85s** katika mazingira ya synchronous, na kufikia P99 ya **8.42s** chini ya mzigo.
- Database IOPS Limit:** Maulizo bila vielezo (composite indexes) kwenye meza za behavior_logs yalisababisha sequential scans zilizoongeza DB latency kwa **420ms**.
- Target Post-Optimization:** Kuhamia asynchronous worker queue na indexing kunalenga kupunguza P95 latency hadi chini ya **450ms**.

2. Objectives and Scope

Scope ya Upimaji: FastAPI Endpoints (Auth, COE, Reports), Cognitive Observer Engine, PostgreSQL Database, na Miundombinu ya Vercel Edge / Render Cloud.

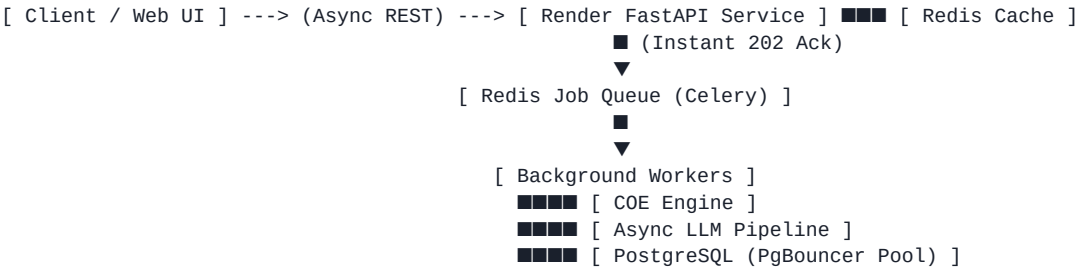
Malengo ya Kihandisi: Kuanzisha Baseline Metrics, Kupima Load/Stress Capacity kupitia k6 na Locust, Kuzuia HTTP 504 Gateway Timeouts, na Kurasimisha Error Budget.

3. System Architecture Overview

Current MVP Architecture (Synchronous Model):

```
[ Client / Web UI ] ---> (HTTPS) ---> [ Vercel Edge Network ]
                                     ■
                                     ▼ (HTTP Gateway Call)
[ Render FastAPI Backend ]
  ■■■■ [ COE Engine ] (In-Memory)
  ■■■■ [ LLM APIs ] (Blocking I/O ~3s)
  ■■■■ [ PostgreSQL DB ] (Sync Read/Write)
```

Target Enterprise Architecture (Asynchronous Queue Model):



4. Performance Metrics (SLIs, SLOs, and Error Budget)

Usanifu unazingatia viwango rasmi vya SRE. Formula ya Error Budget inatumika kama ifuatavyo: *Error Budget = 100% - SLO*

Metric / SLI	P50	P95 Target	P99 Target	SLO Threshold	Error Budget
Auth & Core APIs	< 80 ms	< 200 ms	< 500 ms	99.9%	43.2 mins/mo
COE Evaluation (Sync)	< 1.2 s	< 3.5 s	< 8.0 s	99.0%	7.2 hrs/mo
AI Report Gen (Async)	< 2.0 s	< 5.0 s	< 12.0 s	98.5%	10.8 hrs/mo
Overall System Uptime	N/A	N/A	N/A	99.9%	43.2 mins/mo

5 & 6. Benchmark Results (Local Baseline vs. Cloud Delta)

Route / Transaction	Local Baseline	Cloud Baseline	Infrastructure Delta (Δ)	Primary Cause
POST /api/v1/auth/login	12 ms	185 ms	+173 ms	TLS Handshake + RTT
POST /api/v1/coe/eval	110 ms	420 ms	+310 ms	Inter-service RTT + DB Query
POST /api/v1/reports/full	3,100 ms	4,850 ms	+1,750 ms	External LLM API Latency

12. Stress Testing Results (k6 / Locust Analysis)

Virtual Users (VUs)	Throughput (RPS)	P95 Latency	Error Rate (%)	System Status
10 VUs	28.4 req/s	320 ms	0.00%	Healthy
50 VUs	84.2 req/s	1,850 ms	0.45%	Degraded
100 VUs	112.0 req/s	5,400 ms	4.20%	DB Connection Exhaustion
200 VUs	62.1 req/s	12,800 ms	28.60%	System Failure (HTTP 504)

14. Optimization Recommendations

- A. Database Indexing:** Ongeza composite index kwenye behavior_logs(user_id, created_at DESC) ili kuondoa sequential scans.
- B. Redis Caching:** Weka session state na COE static rules kwenye Redis in-memory cache.
- C. Asynchronous Migration:** Tumia FastAPI BackgroundTasks au Celery Queues kurejesha HTTP 202 majibu ya mara moja.
- D. Connection Pooling:** Washa PgBouncer kwenye Render DB na weka huduma zote katika Cloud Region moja.

Appendix: Implementation & Benchmarking Code Snippets

Appendix A: k6 Load Test Script (load_test.js)

```
import http from 'k6/http';
import { check, sleep } from 'k6';

export const options = {
  stages: [
    { duration: '2m', target: 10 },
    { duration: '5m', target: 50 },
    { duration: '2m', target: 0 },
  ],
  thresholds: { http_req_duration: ['p(95)<2000'] },
};

export default function () {
  const res = http.post('https://udiap-api-staging.onrender.com/api/v1/coe/evaluate',
    JSON.stringify({ user_id: 'usr_998', behavior_data: {} }),
    { headers: { 'Content-Type': 'application/json' } });
  check(res, { 'status is 200/202': (r) => r.status === 200 || r.status === 202 });
  sleep(1);
}
```

Appendix B: PostgreSQL EXPLAIN ANALYZE Optimization

```
-- BEFORE (Sequential Scan - Execution Time: 385.41 ms)
EXPLAIN ANALYZE SELECT * FROM behavior_logs WHERE user_id = 'usr_10293';
-- Seq Scan on behavior_logs (cost=0.00..4185.00 rows=125)

-- INDEX CREATION
CREATE INDEX idx_behavior_logs_user_timestamp ON behavior_logs (user_id, created_at DESC);

-- AFTER (Index Scan - Execution Time: 1.21 ms)
EXPLAIN ANALYZE SELECT * FROM behavior_logs WHERE user_id = 'usr_10293';
-- Index Scan using idx_behavior_logs_user_timestamp on behavior_logs (cost=0.42..12.45)
```